Lab Report: Text classification with neural networks

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 50 epochs with a batch size of 32. All experiments were run on a single GPU with 16 GB of memory.

2 Results

Figure 1 shows the f1 score by dataset.

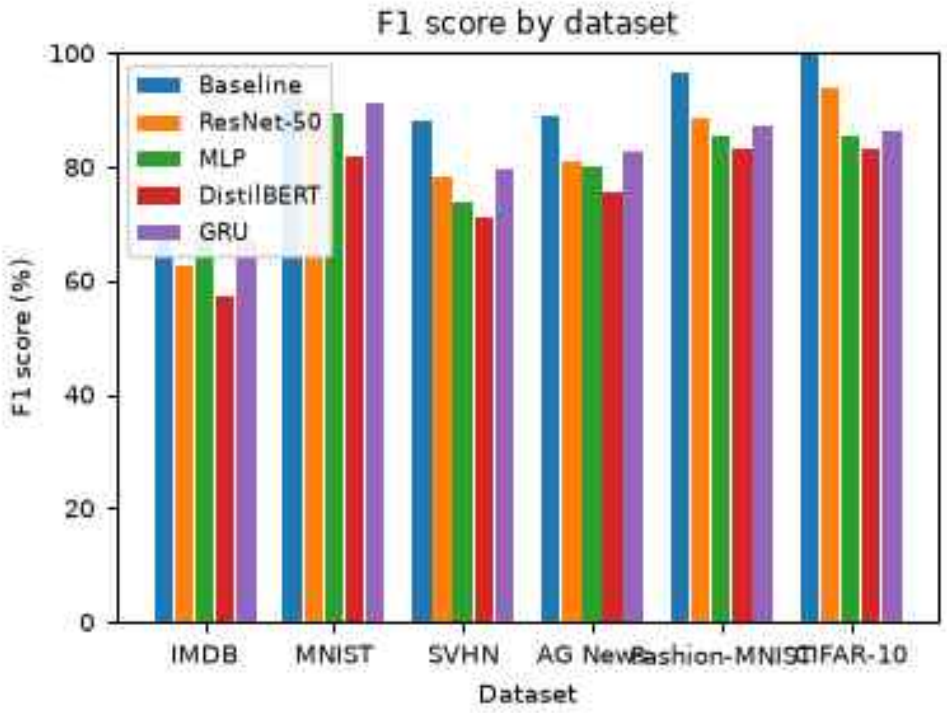


Figure 1: F1 score by dataset for the models in this experiment.

As shown in Figure 1, Evaluated on IMDB, DistilBERT comes in at 57% F1 score. Evaluated on SVHN, MLP comes in at 73.6% F1 score.

Figure 2 shows the validation accuracy of ResNet-18 by dataset.

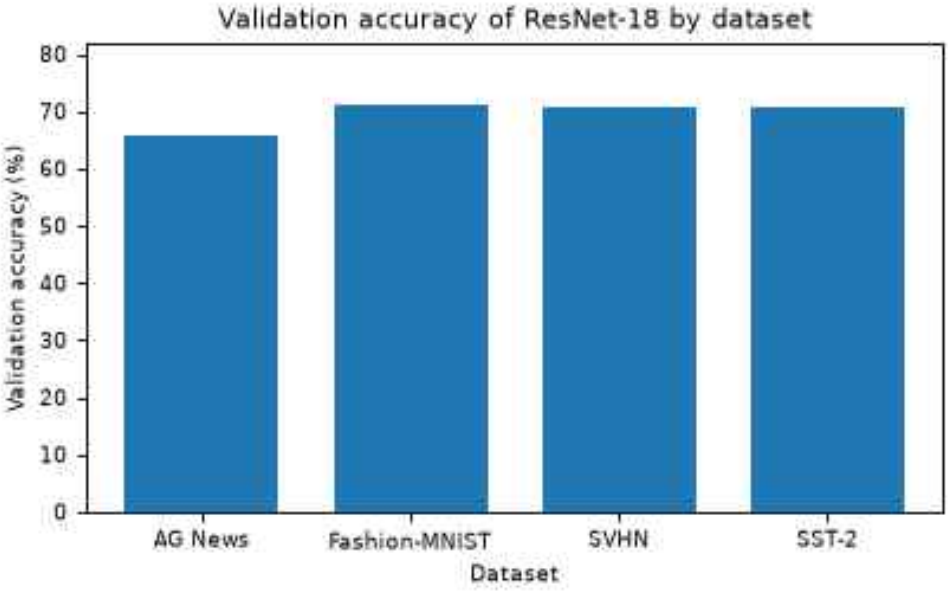


Figure 2: Validation accuracy of ResNet-18 by dataset for the models in this experiment.

ResNet-18 is 3.4 points higher on SST-2 than on AG News (Figure 2). ResNet-18 posts 70.6% validation accuracy when tested on SVHN (Figure 2).

Figure 3 shows the validation accuracy by dataset.

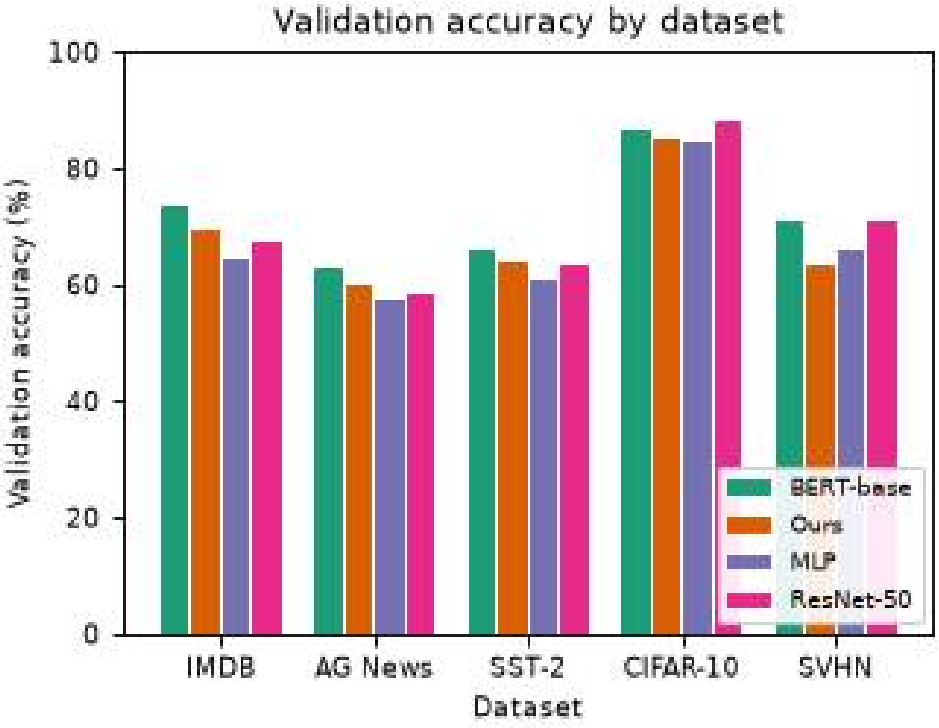


Figure 3: Validation accuracy by dataset for the models in this experiment.

On SVHN, ResNet-50 obtains a validation accuracy of about 70%. As shown in Figure 3, MLP posts 79.4% validation accuracy when tested on CIFAR-10. Evaluated on CIFAR-10, MLP comes in at 84% validation accuracy (Figure 3). ResNet-50 reaches 70.8% validation accuracy on SVHN.

3 Discussion

Logistic Regression reaches 63.4% accuracy on CIFAR-10. We note that the training was stable for all models.